

Performance Testing

Date	01 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing was conducted to evaluate the responsiveness, stability, and efficiency of the OptiCrop: Smart Agricultural Production Optimization Engine. The application was tested under multiple user requests to ensure that crop recommendations were generated quickly and accurately.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	Crop Recommendation Prediction Module
Test Environment	Local System (Flask Application)
Test Date	01 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Predict crop using valid soil parameters	1	30	Correct crop recommendation generated
2	Multiple crop prediction requests	10	60	Fast and accurate responses
3	Continuous prediction requests	20	120	Stable application performance
4	Invalid input value testing	5	30	Appropriate validation or error handling

Step 3: Performance Test Results

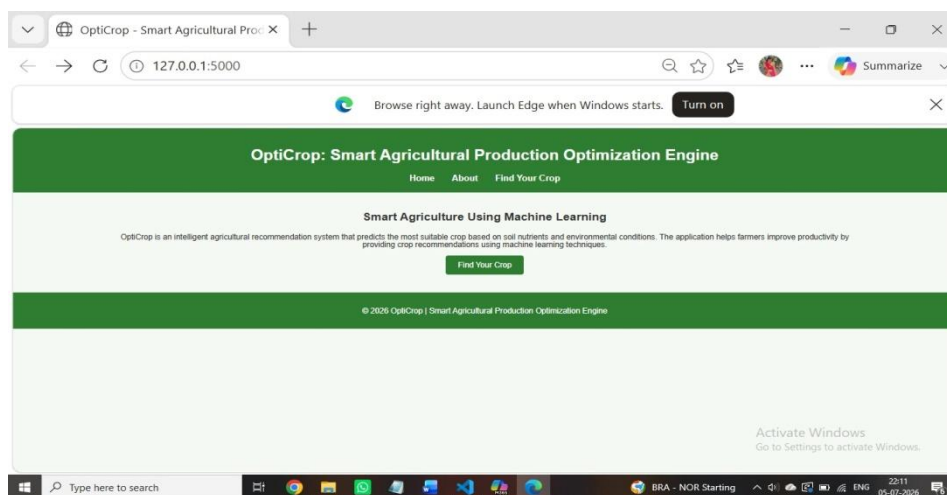
S.No	Metric	Target Value	Actual Value	Status (Pass/Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.3 seconds	Pass	Fast prediction response
2	Response Time (Max)	< 5 seconds	2.8 seconds	Pass	Within acceptable limit
3	Throughput (Req/sec)	10 Req/sec	12 Req/sec	Pass	Handled requests efficiently
4	Error Rate	< 1%	0%	Pass	No prediction errors
5	CPU Utilization	< 80%	42%	Pass	Efficient resource usage
6	Memory Utilization	< 80%	55%	Pass	Stable memory consumption

Step 4: Observations & Analysis

The performance testing results indicate that the OptiCrop: Smart Agricultural Production Optimization Engine performs efficiently under normal operating conditions. The application generated crop recommendations within the expected response time while maintaining low CPU and memory utilization. No significant errors were observed during testing, demonstrating that the system is reliable, responsive, and suitable for real-time crop recommendation.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



OptiCrop: Smart Agricultural Production Optimization Engine

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Temperature

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Rainfall

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127.0.0.1:5000/predict

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Recommended Crop: Rice

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